

Rensselaer Polytechnic Institute
Department of Electrical, Computer, and Systems Engineering
ECSE-4040 - Digital Electronics
Spring Term 2026

Lab. 5-CMOS Circuits and RC Delay

Issued: 2/18/2026

Due: 2/20/2026

Given

Below is a table of process parameters for a generic 0.13 μm technology.

0.13 μm Process technology parameters

Typical Technology Parameters	Symbol	NMOS	PMOS
Gate thickness	t_{ox} [Å]	22	22
Poly doping level	N_D [cm^{-3}]	3×10^{20}	—
Poly doping level	N_A [cm^{-3}]	—	3×10^{20}
Substrate doping level	N_A [cm^{-3}]	3×10^{17}	—
N-well doping level	N_D [cm^{-3}]	—	3×10^{17}
Number of surface state charges	N_{ss} [cm^{-2}]	6×10^{11}	6×10^{11}
Electron Mobility	μ_n [cm^2/Vs]	270	—
Hole Mobility	μ_p [cm^2/Vs]	—	70
Critical Field x Length	$E_C L$ [V]	0.6	2.4
Saturation velocity	v_{sat} [cm/s]	8×10^6	8×10^6
Gamma	γ [$\text{V}^{1/2}$]	0.2	0.2
Oxide capacitance	C_{ox} [F/cm^2]	1.6×10^{-6}	1.6×10^{-6}
Threshold voltage	V_T [V]	0.4	− 0.4
2 x Fermi voltage	$2 \phi_F $ [V]	0.88	0.88
Channel length modulation	λ [V^{-1}]	0.2	0.2

Question 1: Based on Problem 4.14 (D.A. Hodges, H.G. Jackson, and R.A. Saleh, *Analysis and Design of Digital Integrated Circuits*, 3rd Ed., McGraw-Hill, 2004.)

Objectives:

- CMOS inverter design and characterization

Design and characterize propagation delays for a CMOS inverter using the textbook 130nm process parameters with the following V_S points. Adjust PMOS width as needed to achieve the required V_S . Use $R_{ON(NMOS)} = 12.5k$, and $R_{ON(PMOS)} = 30k$, when $W/L = 1$.

- $V_S = 0.6 V_{DD}$
- $V_S = 0.4 V_{DD}$
- Compute and compare τ_{PHL} and τ_{PLH} between part (a) and part (b) with a 1pF load capacitor, using the given R_{ON} values above modified by your calculated W_P .
- Compute and compare τ_{PHL} and τ_{PLH} between part (a) and part (b) with no load capacitor, where C of a $W=130nm$, $L=130nm$ MOSFET = 1fF, using the given R_{ON} values above modified by your calculated W_P .

Question 2: Based on Problem 5.1 (D.A. Hodges, H.G. Jackson, and R.A. Saleh, *Analysis and Design of Digital Integrated Circuits*, 3rd Ed., McGraw-Hill, 2004.)

Objectives:

- Function implementation in CMOS

Design static CMOS gates that have the following outputs. You may assume you have the availability of true complement forms of each signal for each input (Ex: Both A and A' are available without an additional inverter). Design each function as a single gate, where one part of your CMOS circuit does not serve as an input for any other section. Device sizing is not required for this problem.

HINT: Remember that NMOS turn on when they receive a high voltage input. Derive the pull-down circuit equation **Out'** for positive inputs. PMOS turn on when they receive low voltage input. Derive the pull-up equation **Out** for all inverted inputs.

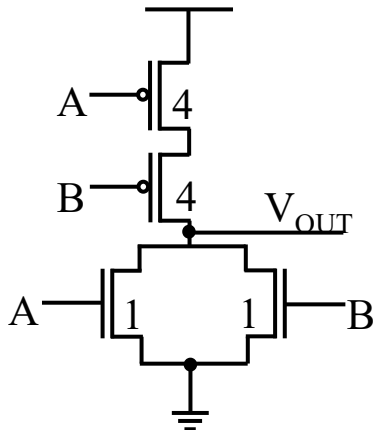
- a. $Out = ABC + BD$
- b. $Out = AB + \overline{A}C + BC$
- c. $Out = \overline{A + B + CD} + A$

Question 3: Based on Problem 5.7 (D.A. Hodges, H.G. Jackson, and R.A. Saleh, *Analysis and Design of Digital Integrated Circuits*, 3rd Ed., McGraw-Hill, 2004.)

Objectives:

- CMOS inverter design and characterization

Consider a 2-input NOR gate where $W_P = 4$ and $W_N = 1$.



- Calculate V_S when only input A is switching.
- Calculate V_S when only input B is switching.
 - NOTE: V_{BS} will affect V_{TP} .
- Calculate V_S when both inputs are tied and switching simultaneously.
- Plot all three VTCs in SPICE.
- Compare your calculated vs simulated results.

Report Submission:

Submit your work on LMS under **Week 6/Lab 5**.

- Show your work.
- Include images of your hand calculations with work and final results.
- Include screenshots of LTspice with all circuit schematics with all output plots, **clearly showing all hand calculated parameters**.
- Due date: Friday at 11:59pm.